

Alt Mobility - Data Analysis Summary

1. SQL Queries and Visual Analysis

The following SQL queries were written and executed using MySQL Workbench via Jupyter Notebook to analyze Alt Mobility's data.

1. Order Status Distribution:

```
SELECT order_status, COUNT(order_id) FROM customer_orders GROUP BY order_status;
```

-> Shows how many orders fall under each status (pending, delivered, etc.).

[Plot: Bar Chart]

2. Total Sales Over Time:

```
SELECT DATE_FORMAT(order_date, '%Y-%m'), SUM(order_amount) FROM customer_orders GROUP BY 1;
```

-> Monthly trend of total revenue.

[Plot: Line Chart]

3. Payment Status Analysis:

```
SELECT payment_status, COUNT(*) FROM payments GROUP BY payment_status;
```

-> Pie chart showing success vs. failed payments.

4. Repeat Customer Analysis:

```
SELECT customer_id, COUNT(order_id) FROM customer_orders GROUP BY customer_id HAVING COUNT(order_id) > 1;
```

-> Distribution of customers who made multiple purchases.

5. Customer Segmentation:

```
SELECT customer_id, SUM(order_amount) FROM customer_orders GROUP BY customer_id;
```

-> Bar chart showing top 20 customers by spend and histogram of spend distribution.

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2. Power BI Visualization - Customer Retention Matrix

The matrix chart below shows cohort-based customer retention using Power BI.

Alt Mobility

Customer Retention Cohort matrix

Year	Month 0	Month 1	Month 2	Month 3	Month 4	Month 5	Month 6	Month 7	Month 8	Month 9	Month 10	Month 11	Month 12
2020	100.00%	1.36%	1.61%	3.69%	2.29%	2.58%	1.69%	1.78%	1.76%	1.41%	1.53%	2.41%	1.53%
2021	100.00%	2.89%	2.85%	2.66%	3.39%	2.22%	1.74%	2.72%	2.59%	2.01%	2.31%	5.55%	2.20%
2022	100.00%	2.45%	15.91%	7.05%	2.00%	4.55%	16.82%	3.86%	1.97%	17.73%	5.61%	2.73%	2.88%
2023	100.00%	1.17%	1.02%	1.17%	1.02%	3.26%	2.10%	3.15%	1.22%	1.14%	1.22%	1.07%	2.14%
2024	100.00%	1.68%	3.08%	1.21%	1.21%	0.91%	1.33%	2.66%	2.24%	0.98%	1.03%	0.56%	0.56%
Total	100.00%	9.71%	4.61%	5.04%	3.00%	2.66%	2.80%	5.67%	5.24%	2.55%	3.13%	3.07%	9.00%

The **Cohort Matrix** is a powerful retention dashboard that helps Alt Mobility (or any business):

- Understand when and why customers return
- Evaluate performance of acquisition channels
- Identify drop-off points in the customer journey

In this matrix:

Month 0 is always 100% which means that everyone in the cohort made a purchase in their first month.

Month 1, 2, etc. shows what % of those original customers made additional purchases.

A **sharp drop** from **100% to 30% to 10%** which means **poor retention**.

This is built by creating CohortMonth, CohortIndex columns, and calculating RetentionRate using DAX.

Insights from the Cohort Matrix:

- Month 0 is always 100%, meaning all customers made their first purchase then.
- Retention drops significantly by Month 1 and 2 (e.g., from 100% to 10-30%).
- 2022 had slightly better retention, suggesting stronger engagement or promotions.

Business Recommendations for Alt Mobility:

- Improve onboarding experience and first-month follow-ups.
- Introduce loyalty points or discounts for returning customers.
- Use personalized messaging or retargeting to re-engage dormant users.
- Analyze high-retention cohorts and replicate their success patterns

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